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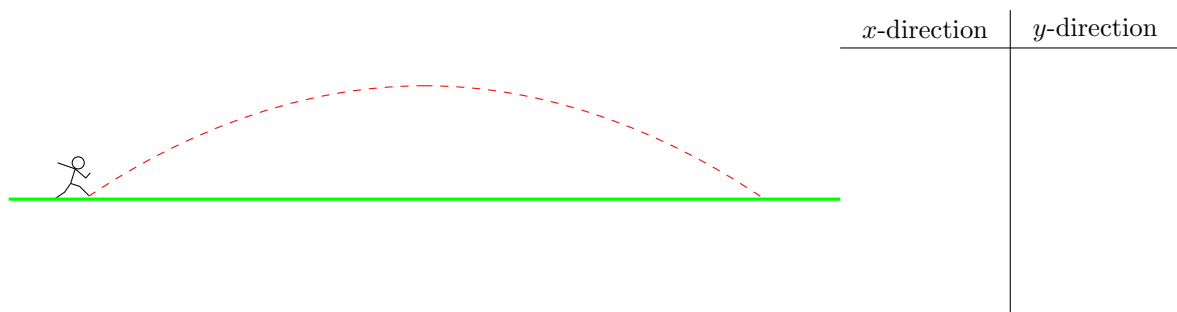
Sub Plans Practice Problem

Instructions Recall that yesterday we learned how to put all of the pieces of our kinematic equations (“*Old Faithful*”, “*Big Chalupa*”, and “*Ain’t Got No Time*”) as well as trigonometry to solve projectile motion problems. Here is another practice problem.

Try to solve the problem on your own or with a partner. The answers are on the back for you to check your progress. I have also included some hints for each part of the problem if you get stuck. This problem is due at the end of the period.

Problem At kickoff, an NFL player kicks a football with an initial resultant velocity of 25 m/s at a direction of 32° above horizontal

- (a) Label the diagram of this situation and write down your knowns and unknowns T-chart. (Remember, only x and y numbers go in the T-chart. Resultants and angles do not.)



- (b) Find the x - and y - components of the ball’s initial velocity. (Draw a triangle and use trig.)
- (c) How long is the ball in the air? (Pick an equation and use either the x -numbers or the y -numbers from your T-chart.)
- (d) How far in the x -direction did it travel?
- (e) What is the football’s maximum height?

Hints

- (a) At this point of the problem, there should only be two things in the T-chart chart: the two accelerations (0 m/s^2 and -9.8 m/s^2)
- (b) This is a Type-II Problem. Check your notes or the front equation board.
- (c) First, make sure to include the answers for part (b) in the T-chart as the v_i 's. You also know that $v_f = 0$ at the top of this motion. So, in the y -direction, you know v_i , v_f , and a . Use Old Faithful to find t . Don't forget that you need to double it!
- (d) Make sure to add the t that you found in part (c) to both sides of the T-chart. In the x -direction, you now know v_i , a , and t . You should be able to use Big Chalupa to find d . Don't forget that $a = 0$ in the x -direction.
- (e) You should only use y numbers to solve this. It's probably easiest to using Ain't Got No Time for this part

Answers

- (a) label the initial velocity vector and the angle.
- (b) $v_{ix} = 21.20 \text{ m/s}$; $v_{iy} = 13.25 \text{ m/s}$
- (c) $t = 2.70 \text{ s}$
- (d) $d_x = 57.32 \text{ m}$
- (e) $d_y = 8.95 \text{ m}$